

1) Nutritional assessment: why & sources
• Purpose: determines nutrition status in ill and healthy people; tested in hospital environment. • A person may enter hospital in good nutrition status then become malnourished; nutrition status often deteriorates during hospitalization. • Illness can ➡ appetite loss; interfere w/ chewing or swallowing; alter digestion or nutrient absorption; alter metabolism or excretion. • One, all, or any combination of these effects may occur during illness and alter nutrition status. • Skilled interview is important in many steps of patient care; information from many sources must be gathered. • Sources: anthropometric measurements, laboratory tests, physical examination, historical data.

2) Anthropometrics
• Physical measurements giving indirect assessment of body composition. • Most useful when repeated periodically and compared with previous measurements to show change in status. • Height and weight are most used; height may not be critical but helps estimate desirable weight and interpret other assessment data. • Other measures: mid-arm circumference, various fat-fold circumference, and head circumference.

3) BMI basics + limitations	
Point	High-yield note
Formula	BMI = BW (kg) / height (m²). For children, BW and height must be in kg and meters.
Clinical use	Helps doctor decide whether child weight is appropriate for height.
Growth	Recently recommended as additional growth measurement. Old weight-for-stature had limited value: boys <11.5 y, girls <10 y only.
Why useful	Better indicator of body “fatness” and potential weight problems.
Risk prediction	In older children/teens, strong correlation between BMI and BF; high BMI usually means high BF and future weight problems.
Prevention	Early identification ➡ careful BF monitoring and potential prevention of adult obesity by eating/exercise changes.
Interpretation limits	Consider race, ethnicity, muscularity, age, sex, other factors. BMI overestimates BF in very muscular persons and underestimates BF after body mass loss, e.g., elderly.

4) Laboratory tests	
Marker / issue	Meaning in nutrition assessment
Plasma albumin	Mild/moderate cases: albumin is ↓. Because albumin is abundant and not broken down quickly, level changes slowly; useful indicator of prolonged protein depletion.
Urinary urea	↓ urinary excretion d/t ↓ protein intake.
3-methylhistidine	↑ urinary 3-methylhistidine reflects muscle breakdown.
Creatinine excretion	↓ excretion reflects atrophy of skeletal muscle mass during malnutrition.
Common association	Iron deficiency anemia commonly assoc. w/ malnutrition.
Kwashiorkor	Characterized by ↓ albumin 10-25 gm/l, ↓ transferrin, ↓ essential amino acids, and ↓ total lymphocyte count.
Important note	Lab tests add dimension, but many factors influence them; no single test alone is reliable. Combine lab tests with anthropometrics and physical findings.

5) Dietary assessment methods
• Dietary evaluation is an important part of nutritional assessment; skilled interview is important. • 24-hour recall: patient describes food eaten in previous 24 hr or typical day. Ask: usual wake time, first meal time, food types; mandatory to ask about amount and preparation method. Not suitable in elderly because memory may be limited. • Food frequency list: asks how often food is eaten in a day or week. • Food records: patient records intake for a specified period, usually 3 or 7 days; in hospital, nursing staff helps provide information.

6) Obesity: intro + definition
• Obesity is a complex disease involving excessive body fat; not just a cosmetic concern. • It is a medical problem that ↑ risk of heart disease, diabetes, high blood pressure and certain cancers. • It is the most prevalent form of malnutrition. • Definition: excess BF; surrogate marker for BF content is BMI. • Clinical terms: BMI 25-29.9 kg/m² = overweight; BMI >30 kg/m² = obese. NIH also lists obesity as 30.0-39.9 kg/m² and morbid obesity as ≥40.0.

7) BMI classification table		
• BMI is a simple weight-for-height index used to classify underweight, overweight and obesity in adults; defined as kg/m².		
Classification	BMI	Disease risk
Underweight	<18.5	-
Normal weight	18.5-24.9	-
Overweight	25.0-29.9	High
Obesity class I	30.0-34.9	Very high
Obesity class II	35.0-39.9	Very high
Obesity class III	≥40.0	Extremely high

*Disease risk refers to type 2 diabetes, hypertension and CVD. ↑ disease risk is assoc. w/ ↑ morbidity.

8) Epidemiology
• Age: obesity can occur at any age and generally ↑ with age. • Sex: women generally have higher obesity rate than men d/t pregnancy and menopause. • Genetic factor: familial tendency; obesity runs in families. • Socio-economic status: clear inverse relationship with obesity. • Physical inactivity: sedentary lifestyle, particularly sedentary occupation. • Alcohol: consumption ↑ risk of obesity. • Human habits: ↓ sleep quality ↑ risk. • Developed countries: obesity prevalence in children and adults is rapidly ↑. • Smoking: smokers weigh somewhat less than ex-smokers; never-smokers fall between the two.

9) Risk factors	
Group	Risk factors
Lifestyle / diet	Excess intake diet (most common); sedentary lifestyle; stress; insufficient sleep; alcohol consumption.
Smoking	Smoking cessation because smoking suppresses appetite.
Endocrine	Hypothyroidism; Cushing’s syndrome; growth hormone deficiency; insulinoma.
Drugs	Corticosteroids; antidepressants.
Female-related	Pregnancy and menopause.
CNS	Hypothalamic damage, e.g., trauma or tumor.
Genetic syndromes	Genetic syndromes assoc. w/ hypogonadism, e.g., Prader-Willi syndrome and Laurence-Moon-Biedl syndrome.

10) C/P	
Area	C/P
Adults	Excess BF accumulation; breathlessness; ↑ sweating; snoring; inability to cope with sudden physical activity; fatigue; back and joint pains; double chin.
Children	Tall stature in some; abdominal striae; obesity of extremities; ↑ adipose tissue in mammary tissue in boys; large pubic fat pad; early puberty.

11) COMP	
COMP	Details
CVD	Hypertension; atherosclerosis.
Metabolic	Type 2 diabetes.
Respiratory	Obstructive sleep apnea.
MSK / mental	Osteoarthritis; depression.
Cancers	↑ risk of colon, liver, pancreas, gallbladder, breast, uterus and prostate cancers.

12) Assessment + Dx criteria	
• Most widely used criteria: BMI, RW, skinfold thickness, WHR, others.	
Method	Criterion / note
BMI	Most widely used; obesity Dx when BMI is 30 or higher. Formula: BW(lb)/height(in²)×703 OR BW(kg)/height(m²). Reasonable estimate of BF but does not directly measure BF. Muscular athletes may fall in obesity BMI category even if muscular.
BMI limits	Age, sex, ethnicity and muscle mass influence BMI-BF relation. BMI does not distinguish excess fat, muscle, or bone mass, and gives no indication of fat distribution; still widely used indicator of excess weight.
RW	RW = BW(kg) / desirable weight ×100. Obese when RW ≥120%.
Skinfold	Rapid, non-invasive, using calipers e.g., Harpenden. Sites: mid-triceps, biceps, subscapular, suprailiac. Sum should be <20 mm boys and <30 mm girls. Dx slide: triceps normal thickness 20 mm men, 30 mm women.
WC	Metabolic complication risk: men >102 cm, women >88 cm.
WHR	High WHR indicates abdominal fat accumulation: men >0.9, women >0.85.
Others	Slide lists “Others” among assessment criteria.

13) Situation in Egypt
• Prevalence of obesity in adults is reported as very high in Egypt, particularly among women. • Overall prevalence of central obesity among Egyptian adults by two indicators: WC 24.1% and WHR 28.7%. • These figures are relatively high considering the association of central obesity with morbidity and mortality. • Many general-obesity studies show significantly higher prevalence among females than males. • Possible sociocultural factors in Egyptian communities: high unemployment, restricted outdoor activities, high illiteracy rate among females.

14) TTT goals + diet + smoking + activity
• Goal of obesity TTT: reach and stay at a healthy weight. • Diet: low-caloric diet; reduction of saturated fats, trans fats and cholesterol. • National Cholesterol Education Program suggests no more than 30% of calories should come from fats. • Smoking cessation: avoid smoking hazards and achieve healthy lifestyle. • Physical activity: significant weight loss can be achieved in many obese persons with ↑ physical activity.

15) Exercise as TTT	
Exercise point	Details
Recommendation	American College of Sports Medicine and Centers for Disease Control recommend every adult accumulate 30 min or more moderate-intensity physical activity on most, preferably all, days/week.
BW / BF	Maintenance of ↓ BW and BF content; prevention of weight/fat regain.
Blood pressure	↓ SBP and DBP; control of hypertension.
Heart	↓ resting heart rate; ↑ stroke volume; ↑ cardiac output.
Lipids	↓ LDL, VLDL, triglycerides, free fatty acids; ↑ high-density lipoproteins.

16) Prevention + therapy mindset
• Good way to monitor diet: keep detailed food diary to know exact calories consumed/day and where calories come from. • Changing habits is hard; person must be ready. Check: is this the right time? ready to make and stay on a plan? family/friends support? first steps known? • Becoming healthier and staying that way is a lifelong effort. • Nutritional education about healthy diet.

17) Practical diet / daily activity points	
Point	Details
Meal size	Keep most meals under 400 calories. Keeping meals same size at regular intervals linked to greater calorie burning after eating, better insulin response, and lower fasting blood cholesterol.
Food quality	Move toward healthier, natural foods; cut processed, rubbish junk foods.
Walking	Daily exercise: 35-40 min brisk walk is doable even for someone overweight.
Water	↑ water between meals, not during meal. Drink 8 glasses/day. Water may speed metabolism; try a glass before meals/snacks and before sweetened drinks or juices.

18) Everyday activity tricks
• Step it up and down: climbing stairs strengthens legs because you lift BW against gravity. • Grocery bags as dumbbells: do not miss the chance for strengthening and calorie burning. Carry groceries balanced with a bag in each hand, even if several trips. • Suitcases: pack two smaller suitcases instead of one big one.

19) Fiber strategy	
Fiber point	Details
Amount	Eat 4 g fiber at every meal. High-fiber diet can lower caloric intake w/o feeling deprived.
Mechanisms	Fiber may slow eating because it requires more chewing; absorbs large water quantities and forms a natural substance filling stomach.
Effects	Produces fullness during and after meal; helps lower cholesterol; gives relief in constipation; speeds passage of food through digestive tract; boosts satiety hormones.
Food note	Buy fruit; it is more useful than vegetables and an easy way to ↑ fiber. One large apple has 5 g fiber, as much as a cup of raw broccoli.

20) Medications + surgery
• Medications: appetite suppressors such as sibutramine; orlistat ↓ absorption of fat. • Surgery: for morbidly obese patient: gastric bypass surgery, gastric banding surgery and vertical sleeve gastrectomy. • These procedures generally produce substantial weight loss in obese patients. • Gastric bypass and vertical sleeve procedures often ➡ rapid remission of type II diabetes mellitus, an important COMP of obesity.

21) Prevention & control	
Prev. / control area	Details
Core idea	Steps to prevent weight gain are the same as steps to lose weight: daily exercise, healthy diet, long-term commitment to watch what you eat and drink.
Regular exercise	Including physical activity in day-to-day activities is important to prevent obesity and other diseases.
Weight monitoring	Regular maintenance of healthy weight and prevention of unnecessary weight gain.
Healthy eating plan	Planning courses of food rich in high-value protein and carbohydrates and low in fat.
Avoidance	Avoid alcohol and smoking.
Health education	Educate children and adults about risk factors, COMP of obesity, and how to avoid them.
Children care	Care of obese children via parents’ observation, control of food proportions, and development of healthy eating habits.
Stress	Avoid stressful events and control stress by medications if necessary to avoid general disease risk.

22) Approved abbreviation legend	
Symbol / abbreviation	Meaning
↑ / ↓ / ➡	increase / decrease / leads to-causes-results in
d/t, min, w/o	due to, minutes, without
assoc. w/, kg/m²	associated with, BMI unit
BMI, BW	Body mass index, body weight
Dx, TTT, COMP, C/P	Diagnosis, treatment, complications, clinical picture
SBP, DBP, LDL, VLDL, CVD	Systolic BP, diastolic BP, lipid abbreviations, cardiovascular disease